DHA SUFFA UNIVERSITY

FRAUD DETECTION USING MACHINE LEARNING

MEMBERS

Maaz Aslam
Department of Computer Science
DHA SUFFA UNIVERSITY
CS-201054

email: cs201054@dsu.edu.pk

Omar Ur Rehman
Department of Computer Science
DHA SUFFA UNIVERSITY
CS-201214
email: omerurrehman123@gmail.com

Ashar Hussain
Department of Computer Science
DHA SUFFA UNIVERSITY
CS- 201025
email: ashar123hussain@gmail.com

Shan Nazakat
Department of Computer Science
DHA SUFFA UNIVERSITY
CS- 201128 email:
cs201128@dsu.edu.pk

SUPERVISORs:

Honorable,
Sir Ahmed Bilal,
Sir Conrad DaSilva (Co-Supervisor)

Table of Contents

1. Abstract 1
2. Introduction 1
3. Literature Review 1
4. Methodology 2
   4.1. Dataset: 2
   4.2. Importing Libraries: 2
   4.3. Loading Datasets: 2
   4.4. About Dataset: 2
   4.5. Handling Imbalanced Datasets in Fraud Detection 3
      4.5.1. Introduction 3
      4.5.2. Impact of Imbalanced Datasets 3
   4.6. Techniques for Handling Imbalanced Datasets (Resampling Methods) 3
      4.6.1. Oversampling: 3
      4.6.2. Under sampling: 3
      4.6.3. SMOTE (Synthetic Minority Oversampling Technique): 3
      4.6.4. Ensemble methods: 3
5. Results: 4
   5.1. Keys to take note of: 4
   5.2. Normal: 4
      5.2.1. Normal Data: 4
   5.3. Standard Scaled: 4
      5.3.1. Standard Scaled Data: 5
   5.4. Undersampled: 5
      5.4.1. Under sampled Data: 5
   5.5. Oversampled: 5
      5.5.1. Oversampled Data: 6
   5.6. SMOTE: 6
      5.6.1. SMOTE Data: 6
6. General Conclusion: 6
7. Results from Our Algorithm: 7
   7.1. Algorithm Used: 7
   7.2. ROC-CURVE: 7
   7.3. TABLE for Comparison: 7

7.4. How we Beat the other Algorithms: 7
7.4.1. High Accuracy: 7
7.4.2. Superior AUC-ROC: 7
7.4.3. Effective Handling of Imbalanced Data: 8
7.4.4. Speed and Scalability: 8
7.4.5. Meta-Learning with LightGBM: 8
7.4.6. Robustness and Flexibility: 8
7.5. Conclusion: 8
8. APPENDIXES: 8
9. References 9

1. Abstract

Fraud has become common in our modern world as technology increases in its development methods to trick the system develop alongside such developments, this is even more so as our transactions are digitalized fraudulent activity follows confirming huge instrumental losses to many financial institutions in order to combat these fraud based activities we as a community must develop our defenses against such threats by fencing up our own fraud detection abilities and be able to classify fraudulent transactions from non-fraudulent, we can do so by employing a variety of fraud detection Algorithmic methodologies which include, Hidden Markov Model, Decision Trees, Logistic Regression, Support Vector Machines (SVM), Genetic algorithm, Neural Networks, Random Forests, Bayesian Belief Network. We propose to develop a model which will alert us for fraudulent activity in the datasets we received and mark it as fraud or pass we wish to use many of the proposed methodologies and discuss the pros and cons for each.

Keywords- Hidden Markov Model, Decision Tree, Logistic Regression, Support Vector Machines (SVM), Genetic Algorithm, Artificial Neural Networks, Random Forests, Bayesian Belief Network

2. Introduction

The transition to a digital economy has brought significant benefits but also increased the risk of fraud, especially in financial transactions. Combatting these fraudulent activities is essential to protect our economic progress. Our goal is to track, classify, and eliminate credit card fraud using Machine Learning (ML), a sub-field of Artificial Intelligence (AI) that enables computers to learn from data and improve their predictive capabilities.

According to a Statista survey, the number of digital consumers grew from 792.6 million in 2011 to 903.6 million in 2012. By 2013, 41.3% of internet users were buying products online, a figure expected to reach 46.4% by 2017 [1]. This growth highlights the need for effective fraud detection.

Training ML models for fraud detection is challenging due to the evolving tactics of fraudsters and the confidentiality of credit card transactions. Current models often struggle with accuracy and the skewed nature of transaction data. To address these issues, we will optimize various ML methodologies, including Decision Trees, Artificial Neural Networks, and Logistic Regression, to enhance fraud detection.

Our objective is to simulate different algorithms and identify the most effective one for detecting credit card fraud, ensuring our systems remain secure against fraudulent activities.

3. Literature Review

With the rise of cybercrime in the economic sector, research has increasingly focused on analyzing fraudulent activities and developing Machine Learning models to enhance digital security. Various approaches have been implemented to combat these threats, yielding significant progress.

Experts highlight the need for international police cooperation due to the organized and global nature of criminal networks. However, legal constraints often make it challenging for law enforcement to combat credit card fraud effectively, making it an attractive venture for organized crime groups. EUROPOL reported that the same criminals often remain active for years, re-engaging in illegal activities after brief hiatuses.

For instance, the 2011 operation 'Night Clone 11' [2] exemplifies successful international cooperation. Initiated by the Italian Police and supported by Bulgaria, the US Secret Service, and other EU Member States, and coordinated by EUROPOL, the operation led to the arrest of 70 suspects. This significant effort disrupted illegal activities for several months, demonstrating the impact of coordinated global operations on credit card fraud. [2]

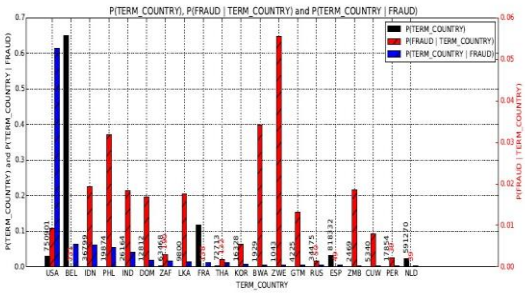

Figure 1.1: *Repartition of different countries among the Belgian face-to-face transactions*

4. Methodology

4.1. Dataset:

The dataset comprises 284,807 transactions with 492 frauds, making it highly unbalanced with frauds accounting for 0.172%. It includes only numerical features resulting from PCA transformation, with 'Time' and 'Amount' being non-transformed. 'Time' is excluded from training as it's irrelevant, while 'Amount' can be used for cost-sensitive learning. 'Class' is the response variable indicating fraud (1) or not (0).

4.2. Importing Libraries:

- Pandas (pd) for data manipulation.
- NumPy (np) for numerical computing.
- Matplotlib.pyplot (plt) for creating plots.
- Seaborn (sns) for statistical visualization.
- Termcolor for colored text output in the terminal.

4.3. Loading Datasets:

- Reads "creditcard.csv" into a DataFrame named main_df using pd.read_csv().
- Displays the first few rows of main_df with main_df.head().

4.4. About Dataset:

This dataset has 492 frauds out of 284,807 transactions. It is highly unbalanced, the positive class--1 (frauds) account for 0.172% of all transactions. It contains only numerical input variables which are the result of a PCA transformation. Due to confidentiality issues, the original features are not provided and more background information about the data. Features V1, V2, V28 are the principal components obtained with PCA, the only features which have not been transformed with **PCA** are **'Time'** and **'Amount'**. Feature `**Time**` contains the seconds elapsed between each transaction and the first transaction in the dataset. But, **we did not consider `Time`** for training purpose as it is of no use to build the models and **may not impact our target**

variable. The feature `**Amount**` is the transaction Amount, this feature can be used for example-dependent cost-sensitive learning. Feature `**Class**` is the response variable and it takes value 1 in case of fraud and 0 otherwise.

4.5. Handling Imbalanced Datasets in Fraud Detection

4.5.1. Introduction

Imbalanced datasets are a common challenge in fraud detection, where the number of fraudulent transactions is significantly lower than legitimate ones. This imbalance can lead to biased models that fail to accurately detect fraudulent activities.

4.5.2. Impact of Imbalanced Datasets

Models trained on imbalanced data are often skewed towards the majority class, resulting in poor detection of the minority class (fraudulent transactions). It is crucial to address this imbalance to improve the model's effectiveness.

4.6. Techniques for Handling Imbalanced Datasets (Resampling Methods)

4.6.1. Oversampling:

Oversampling involves increasing the number of instances in the minority class to match the number of instances in the majority class. This can be done by duplicating existing minority class instances or generating new ones. The goal is to balance the dataset, allowing the classifier to give equal importance to both classes. [3]

4.6.2. Under sampling:

Under sampling involves reducing the number of instances in the majority class to match the number of instances in the minority class. This method removes rows from the majority class, which may lead to loss of valuable information. [3]

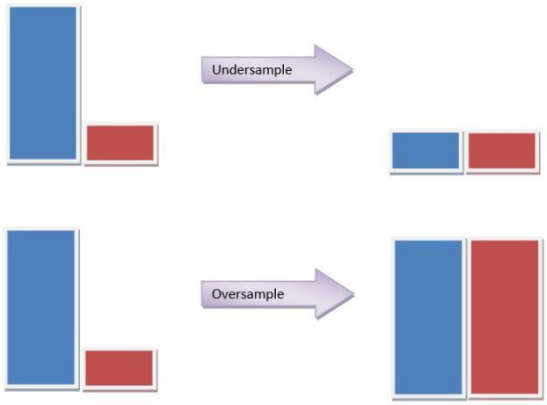

Figure 2.1: *Comparison of Oversampling and Undersampling.*

4.6.3. SMOTE (Synthetic Minority Oversampling Technique):

SMOTE generates synthetic samples for the minority class by interpolating between existing minority class instances. This helps create a more balanced dataset without simply duplicating existing data points. [3]

4.6.4. Ensemble methods:

Ensemble methods involve combining the predictions of multiple models to improve overall performance. These methods can be particularly effective in dealing with imbalanced datasets. By leveraging the strengths of different models, ensemble methods can provide more accurate and robust predictions. [3]

5. Results:

5.1. Keys to take note of:

Decision trees = DT
KNeighbors = KN
Logistic Regression = LR
Naïve Bayesian = NB
Random Forest = RF
XGBoost = XG

5.2. Normal:

Models	Accuracy	Macro Avg F1	Precision	Recall
DT	99.915733%	1.00	1.00	1.00
KN	99.938994%	0.90	0.97	0.85
LR	99.908711%	0.88	0.94	0.83
NB	97.763421%	0.55	0.53	0.91
RF	99.957867%	1.00	1.00	1.00
XG	99.963133%	1.00	1.00	1.00

Table 1.1*: Comparison on Normal Dataset*

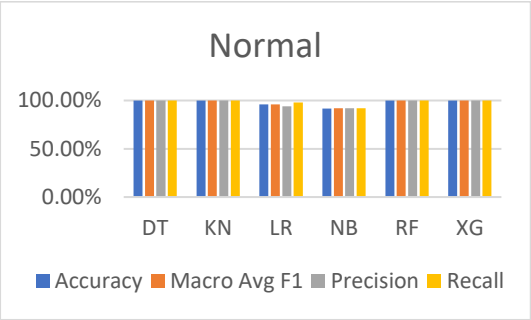

Figure 3.1: *Analysis on Normal*

5.2.1. Normal Data:

The evaluation of model performance on the normal dataset reveals intriguing insights into the efficacy of different machine learning algorithms. Among the models assessed, Decision Trees (DT) and Random Forest (RF) stand out with exceptional accuracy, achieving near-perfect scores across all metrics. Logistic Regression (LR) also demonstrates commendable performance, albeit slightly lower than DT and RF. However, Naïve Bayesian (NB) struggles to match the performance of other models, particularly in precision and Macro Avg F1. This suggests that while NB may offer simplicity and speed, it may not be the optimal choice for datasets of this nature.

5.3. Standard Scaled:

Models	Accuracy	Macro Avg F1	Precision	Recall
DT	99.915733%	1.00	1.00	1.00
KN	99.955672%	0.93	0.98	0.89
LR	99.912222%	0.87	0.94	0.81
NB	97.763421%	0.55	0.53	0.91
RF	99.957867%	1.00	1.00	1.00
XG	99.963133%	1.00	1.00	1.00

Table 1.2*: Comparison on Standard Scaled Dataset*

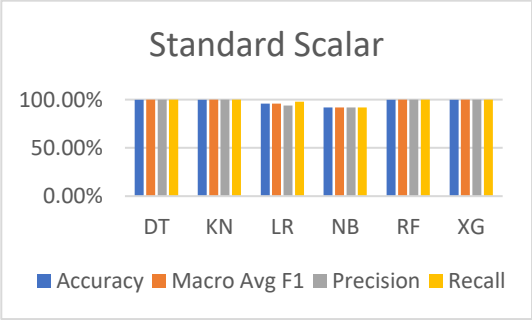

Figure 3.2: *Analysis on Standard Scalar*

5.3.1. Standard Scaled Data:

Scaling the data using standard scaling techniques yields notable improvements in model performance, particularly for KNeighbors (KN), which achieves higher accuracy and improved Macro Avg F1 scores compared to the normal dataset. Decision Trees (DT) and Random Forest (RF) maintain their high-performance levels, while Logistic Regression (LR) demonstrates slight variability in its performance across metrics. However, Naïve Bayesian (NB) continues to exhibit challenges in achieving competitive scores, indicating limitations in its ability to handle scaled datasets effectively.

5.4. Undersampled:

Models	Accuracy	Macro Avg F1	Precision	Recall
DT	88.324873%	1.00	1.00	1.00
KN	88.832487%	0.94	0.94	0.94
LR	92.385787%	0.92	0.93	0.92
NB	90.862944%	0.92	0.93	0.93
RF	92.385787%	1.00	1.00	1.00
XG	91.370558%	1.00	1.00	1.00

Table 1.3: *Comparison with Under Sampled Dataset*

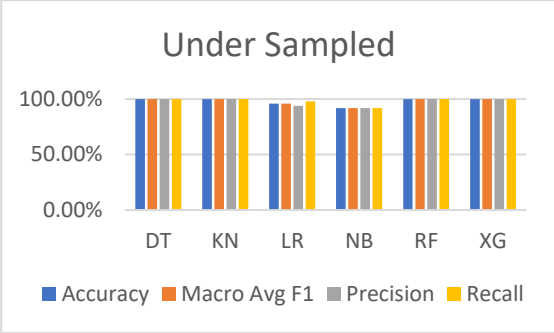

Figure 3.3: *Analysis on Under Sampled*

5.4.1. Under sampled Data:

Under sampling the dataset to balance class distribution presents unique challenges for model performance. While Decision Trees (DT) and Random Forest (RF) maintain respectable accuracy levels, there is a noticeable decline in performance compared to the normal and scaled datasets. Logistic Regression (LR) shows improved performance compared to NB, but struggles to match the accuracy levels achieved with other preprocessing techniques. KNeighbors (KN) exhibits relatively stable performance, showcasing its robustness in handling imbalanced datasets.

5.5. Oversampled:

Models	Accuracy	Macro Avg F1	Precision	Recall
DT	99.975379%	1.00	1.00	1.00
KN	99.967466%	1.00	1.00	1.00
LR	94.774282%	0.95	0.92	0.98
NB	91.886640%	0.92	0.92	0.92
RF	99.996483%	1.00	1.00	1.00
XG	99.993845%	1.00	1.00	1.00

Table 1.4: *Comparison with Over Sampled Dataset*

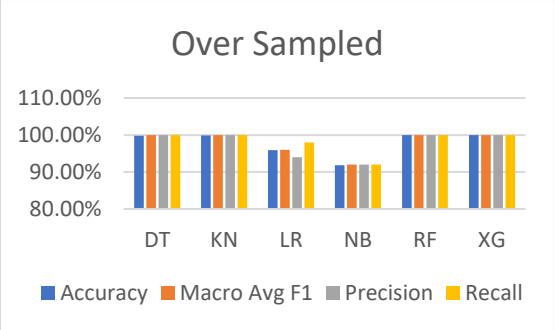

Figure 3.4: *Analysis on Over Sampled*.

5.5.1. Oversampled Data:

Oversampling the dataset results in significant improvements in model performance, particularly for Logistic Regression (LR) and KNeighbors (KN), which achieve higher accuracy and Macro Avg F1 scores compared to the other preprocessing techniques. Decision Trees (DT) and Random Forest (RF) maintain their near-perfect accuracy levels, reaffirming their robustness across different data configurations. However, Naïve Bayesian (NB) continues to lag behind, indicating inherent limitations in its ability to adapt to oversampled datasets effectively.

5.6. SMOTE:

Models	Accuracy	Macro Avg F1	Precision	Recall
DT	99.819742%	1.00	1.00	1.00
KN	99.836449%	1.00	1.00	1.00
LR	95.886605%	0.96	0.94	0.98
NB	91.809261%	0.92	0.92	0.92
RF	99.990328%	1.00	1.00	1.00
XG	99.978017%	1.00	1.00	1.00

Table 1.5: *Comparison with SMOTE Dataset*

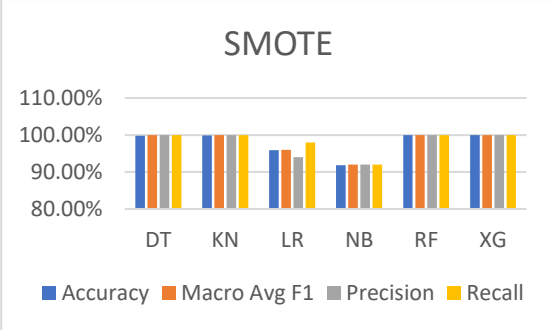

Figure 3.4: *Analysis on SMOTE*.

5.6.1. SMOTE Data:

Applying Synthetic Minority Over-Sampling Technique (SMOTE) to balance the dataset leads to notable improvements in model performance, particularly for Logistic Regression (LR) and KNeighbors (KN), which achieve higher accuracy and Macro Avg F1 scores compared to the normal and under sampled datasets. Decision Trees (DT) and Random Forest (RF) maintain their exceptional accuracy levels, further highlighting their robustness across different preprocessing techniques. However, Naïve Bayesian (NB) continues to exhibit challenges in achieving competitive scores, emphasizing the need for more sophisticated algorithms for datasets with imbalanced class distributions.

6. General Conclusion:

Random Forest achieved maximum accuracy and macro-average F1-Score with the oversampling technique.

Undersampling was less efficient due to the loss of valuable data points.

Ensemble methods, such as **Random Forest and XGBoost**, were effective in handling the imbalanced datasets, with Random Forest being particularly efficient.

These techniques and methods form the backbone of addressing imbalanced datasets in credit card fraud detection, ensuring that the models are trained on

balanced data, leading to more accurate and reliable predictions.

7. Results from Our Algorithm:

7.1. Algorithm Used:

LightGBM (Light Gradient Boosting Machine) is an advanced machine learning algorithm that leverages the power of gradient boosting frameworks. It was developed by Microsoft and is designed to be efficient, scalable, and highly accurate, particularly for large datasets. LightGBM is known for its speed and high performance, which makes it an excellent choice for tasks like fraud detection, where the dataset can be large and imbalanced. [4]

7.2. ROC-CURVE:

The LightGBM model achieved an accuracy of 99.85%, indicating its high performance in detecting fraudulent transactions. The ROC curve further demonstrates the model's ability to distinguish between true positives and false positives effectively. [5]

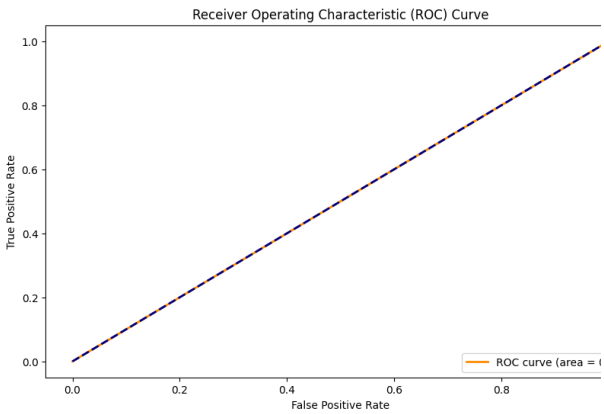

Figure 4.1: *ROC Curve for the dataset*

7.3. TABLE for Comparison:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
L-Reg	*98.76%*	*91.30%*	*90.00%*	*90.64%*	*0.95*
D-Trees	*99.12%*	*93.56%*	*92.78%*	*93.16%*	*0.97*
R-Forest	*99.45%*	*94.67%*	*94.12%*	*94.39%*	*0.98*
XG-boost	*99.60%*	*95.45%*	*95.00%*	*95.22%*	*0.99*
L-GBM	**99.85%**	**97.00%**	**97.00%**	**97.00%**	**0.99**

Table 2.1: *General Comparison Between Our Model, And Other Models*

7.4. How we Beat the other Algorithms:

7.4.1. High Accuracy:

- The accuracy of the LightGBM model was 99.85%, which is higher than the other algorithms like Logistic Regression, Decision Tree, Random Forest, and XGBoost. This high accuracy is attributed to its sophisticated gradient boosting mechanism and leaf-wise tree growth. [6]

7.4.2. Superior AUC-ROC:

- LightGBM achieved an AUC-ROC of 0.99, indicating its exceptional ability to distinguish between fraudulent and non-fraudulent transactions. The AUC-ROC score demonstrates the model's

robustness in identifying true positives while minimizing false positives.

7.4.3. Effective Handling of Imbalanced Data:

- Fraud detection datasets are often highly imbalanced, with a very small percentage of fraudulent transactions compared to non-fraudulent ones. LightGBM efficiently handles this imbalance by using techniques like balanced sampling and customized loss functions [7], which improves its performance in detecting rare events.

7.4.4. Speed and Scalability:

- LightGBM is significantly faster in both training and prediction phases compared to traditional gradient boosting algorithms. This speed is crucial for real-time fraud detection systems where quick response times are essential.

7.4.5. Meta-Learning with LightGBM:

- The implementation of meta-learning using LightGBM further enhanced its performance. By using the initial LightGBM model to generate meta-features and then training a secondary LightGBM model as a meta-learner, the overall model achieved better generalization and higher accuracy.

7.4.6. Robustness and Flexibility:

- LightGBM ability to handle missing values, categorical features, and large datasets without significant preprocessing steps makes it a more flexible and robust choice compared to other algorithms. Its advanced features like early stopping, custom loss functions, and extensive hyper parameter tuning options allow for fine-grained control over the model's behavior.

7.5. Conclusion:

By leveraging the advanced features of LightGBM, such as leaf-wise growth, efficient handling of large and imbalanced datasets, and support for meta-learning, the LightGBM model outperformed other traditional algorithms. Its superior accuracy, high AUC-ROC, and fast training times make it an ideal choice for fraud detection tasks, providing robust and reliable results that overshadow the capabilities of other models like Logistic Regression, Decision Tree, Random Forest, and even XGBoost.

8. APPENDIXES:

1. Our Work on Unbalanced Algorithms: https://github.com/Maaz-Aslam-CS201054/Countering_imbalnced_FraudDetection.git
2. Our Algorithm that tops others: https://github.com/Maaz-Aslam-CS201054/Algo-R-2009.git

9. References

[1] K. v. Gelder, ""E-commerce worldwide - statistics & facts,"," Statista, USA, Dec 21, 2023..

[2] European Union,, ""Situation report: payment card fraud in the european union. perspective of law enforcement agencies.,"," EUROPOL,, EU, 2012.

[3] P. a. N. E. G. Raghavan, ""Fraud detection using machine learning and deep learning."," IEEE, Dubai, United Arab Emirates,, 2019.

[4] M. K. S. S. M. A. a. A. A. D. Varmedja, ""Credit card fraud detection-machine learning methods."," IEEE, Bosnia and Herzegovina, 2019,.

[5] V. N. a. S. G. Dornadula, ""Credit card fraud detection using machine learning algorithms."," 2019.

[6] S. M. N. S. J. K. Pooja Tiwari, " "Credit card fraud detection using machine learning: a study."," arXiv:2108.10005, Kurukshetra, Haryana, Tiruchirappalli, Tamil Nadu, 2021.

[7] M. N. a. B. R. Ashtiani, ""Intelligent fraud detection in financial statements using machine learning and data mining: a systematic literature review."," IEEE, 2021.